Assignment Day 3 (02/03/2026)

Assignment 1

Book and Author (Containment with Array of

Objects + Utility Class)

Design a C++ program to manage a Library System using containment and a separate utility class for operations.

Requirements

Create class Author

Data Members:

- authorName
- country

Member Functions:

- acceptAuthor()
- displayAuthor()

Create class Book

Data Members:

- bookId
- bookTitle
- price
- Contains an object of class Author (Containment)

Member Functions:

- acceptBook()
- displayBook()

This demonstrates a "has-a" relationship (Book has an Author).

Create class PrintToScreen(Utility class)

This class should handle all array operations.

Static Member Functions:

1. acceptBooks(Book b[])
    - Accept details for an array of Book objects.
2. displayBooks(Book b[])
    - Display all book details including author information.
3. searchById(Book b[], int id)
    - Search and display book details by bookId.
    - Display message if not found.
4. searchByAuthor(Book b[], string authorName)
    - Search and display books written by a specific author.

In main() Function

- Create an array of 3 Book objects.
- Call methods from PrintToScreen:
    - acceptBooks()
    - displayBooks()
    - searchById()
    - searchByAuthor()

Solution 1:

Project Location:

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem

File Locations

Header File:

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem\Author.h

```
#pragma once
#include <string>
#include<iostream>
using namespace std;
class Author
{
private:
	string authorName;
	string country;
public:
	Author();
	Author(string authorName, string country);
	string getAuthorName();
	string getCountry();
	void acceptAuthor();
	void displayAuthor();
};
```

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem\Book.h

```
#pragma once

#include<iostream>

#include"Author.h"

using namespace std;


class Book

{

private:

	int bookId;

	string bookTitle;

	float price;

	Author author;

public:

	Book();

	Book(int bookId, string bookTitle, float price, string authorName, string country):
author(authorName, country){

		this->bookId = bookId;

		this->bookTitle = bookTitle;

		this->price = price;

	}

	int getBookId();

	string getBookTitle();

	float getPrice();

	void acceptBook();

	void displayBook();

	Author getAuthor();

};
```

Resource File

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem\author_src.cpp

```
#include"Author.h"
Author::Author() {
	authorName = "";
	country = "";
}
Author::Author(string authorName, string country) {
	this->authorName = authorName;
	this->country = country;
}
string Author::getAuthorName() {
	return authorName;
}
string Author::getCountry() {
	return country;
}
void Author::acceptAuthor() {
	cout << "Enter Author Name: ";
	getline(cin, authorName);
	cout << "Enter Country: ";
	getline(cin, country);
}
void Author::displayAuthor() {
	cout << "Author Name: " << authorName << endl;
	cout << "Country: " << country << endl;
}
```

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem\book_src.cpp

```
#include"Book.h"

Book::Book() {
	bookId = 0;
	bookTitle = "";
	price = 0.0;
}

int Book::getBookId() {
	return bookId;
}
string Book::getBookTitle() {
	return bookTitle;
}
float Book::getPrice() {
	return price;
}
void Book::acceptBook() {
	cout << "Enter Book Id: ";
	cin >> bookId;
	cin.ignore();
	cout << "Enter Book Title: ";
	getline(cin, bookTitle);
	cout << "Enter Price: ";
	cin >> price;
```

```
	cin.ignore();
	cout << "Enter Author Details: " << endl;
	author.acceptAuthor();
}
void Book::displayBook() {
	cout << "Book Id: " << bookId << endl;
	cout << "Book Title: " << bookTitle << endl;
	cout << "Price: " << price << endl;
	cout << "Author Details: " << endl;
	author.displayAuthor();
}

Author Book::getAuthor() {
	return author;
}
```

Client/Main file

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem\client.cpp

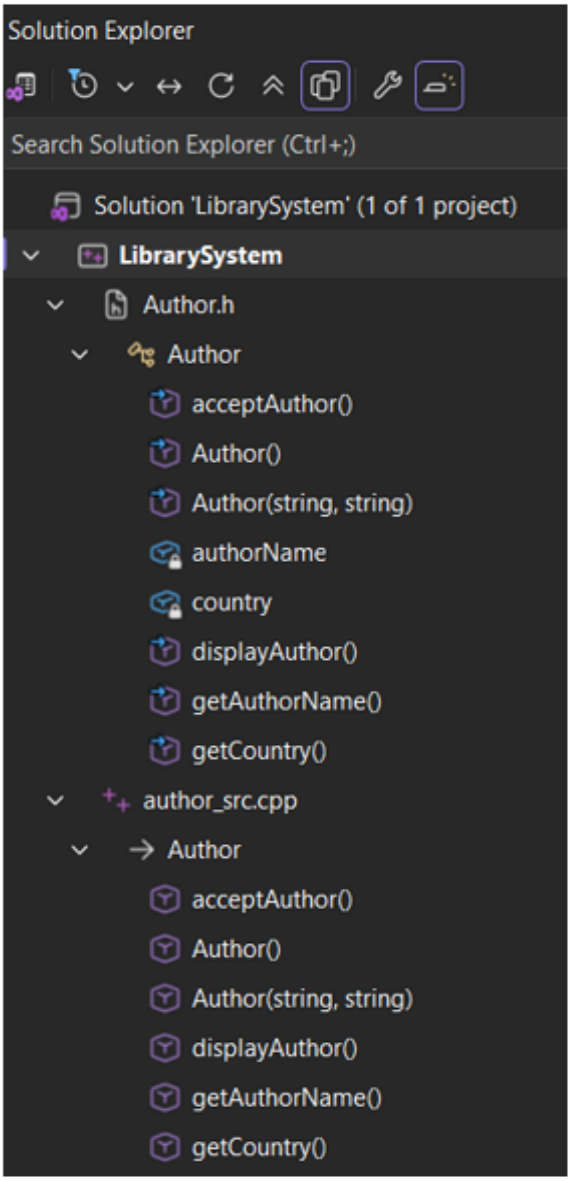
```
#include"Author.h"
#include"Book.h"
using namespace std;

class PrintToScreen
{
public:
	static void acceptBooks(Book books[], int size) {
```

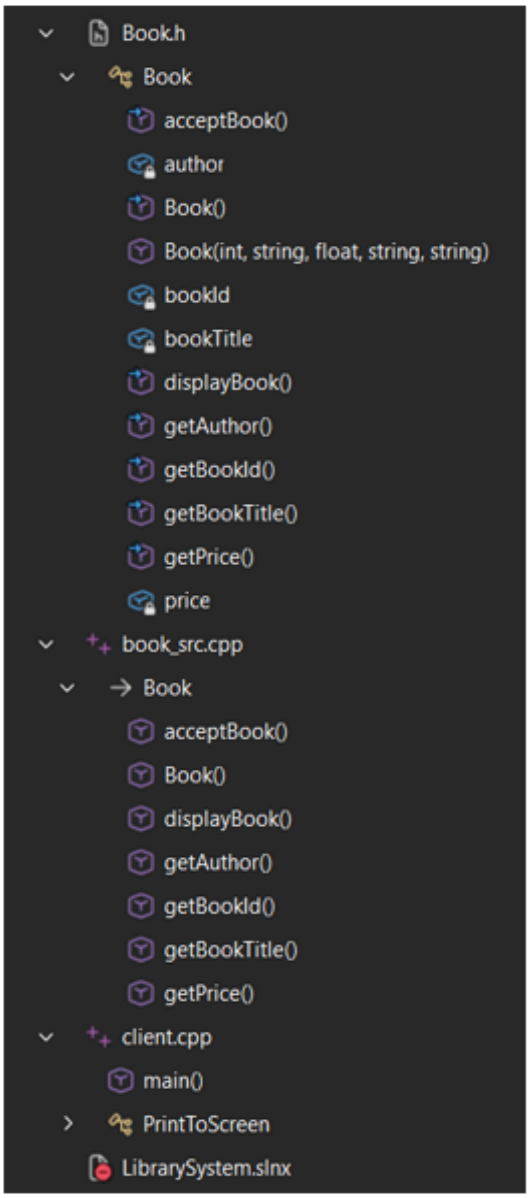
```
        for (int i = 0; i < size; i++) {
                cout << "Enter details for Book " << i + 1 << ":" << endl;
                books[i].acceptBook();
                cout << endl;
        }
}
static void displayBooks(Book books[], int size) {
        for (int i = 0; i < size; i++) {
                cout << "Details of Book " << i + 1 << ":" << endl;
                books[i].displayBook();
                cout << endl;
        }
}
static void searchBookById(Book books[], int size, int bookId) {
        bool flag = false;
        for (int i = 0; i < size; i++) {
                if (books[i].getBookId() == bookId) {
                        flag = true;
                        books[i].displayBook();
                        break; // Exit loop after finding the book
                }
        }
        if (flag == false) {
                cout << "Book with Id " << bookId << " not found." << endl;
        }
}
static void searchByAuthorName(Book books[], int size, string authorName) {
```

```
		bool flag = false;
		for (int i = 0; i < size; i++) {
			if (books[i].getAuthor().getAuthorName() == authorName) {
				flag = true;
				books[i].displayBook();
				break; // Exit loop after finding the book
			}
		}
		if (flag == false) {
			cout << "Book with Author Name : " << authorName << " not found." << endl;
		}
	}
};



int main(){

	Book book[3];
	int choice;
	char wish=' ';
	string authorName="Na";
	int bookId;

	PrintToScreen::acceptBooks(book, 3);
	do
	{
		cout << "\n1.Search Book by Id\n2.Search Book by Author Name. \n\nEnter your choice : 
";
```

```
        cin >> choice;
        cin.ignore();
        switch (choice)
        {
            case 1:
                cout << "Enter Book Id to search : ";
                cin >> bookId;
                cin.ignore();
                PrintToScreen::searchBookById(book, 3, bookId);
                break;
            case 2:
                cout << "Enter Author Name to search : ";
                getline(cin, authorName);
                PrintToScreen::searchByAuthorName(book, 3, authorName);
                break;
            default:
                cout << "Invalid choice." << endl;
                break;
        }
        cout << "Do you want to continue? (y/n) : ";
        cin >> wish;
    } while (wish=='y' || wish=='Y');

    return 0;
}
```

Structure Of Program





}

OUTPUT:

```
Microsoft Visual Studio Debug Console
Enter Author Name: Yashavant Kanetkar
Enter Country: India

Enter details for Book 2:
Enter Book Id: 102
Enter Book Title: Introduction to Algorithm
Enter Price: 2499
Enter Author Details:
Enter Author Name: Thomas H. Cormen
Enter Country: Us

Enter details for Book 3:
Enter Book Id: 103
Enter Book Title: C++
Enter Price: 850
Enter Author Details:
Enter Author Name: Bjarne
Enter Country: US


1.Search Book by Id
2.Search Book by Author Name.

Enter your choice : 1
Enter Book Id to search : 102
Book Id: 102
Book Title: Introduction to Algorithm
Price: 2499
Author Details:
Author Name: Thomas H. Cormen
Country: Us
Do you want to continue? (y/n) : y

1.Search Book by Id
2.Search Book by Author Name.

Enter your choice : 2
Enter Author Name to search : Bjarne
Book Id: 103
Book Title: C++
Price: 850
Author Details:
Author Name: Bjarne
Country: US
Do you want to continue? (y/n) : n

D:\LearnC++\Day5_02032026\Ass1\LibrarySystem\x64\Debug\LibrarySystem.exe (process 5168) exited with code 0 (0x0).
Press any key to close this window . . .
```

Assignment 2

2. Problem Statement: Healthcare Management System

(Hierarchical Inheritance in C++)

Design a C++ program to demonstrate Hierarchical Inheritance using a Healthcare Management System.

Scenario

A hospital manages different types of medical staff such as:

- Doctor
- Nurse
- LabTechnician

All medical staff share common information, but each role has its own specific responsibilities and details.

Requirements

Create a Base Class MedicalStaff

Data Members:

- staffId
- name
- age
- contactNumber

Member Functions:

- acceptStaffDetails()
- displayStaffDetails()

Create Derived Class Doctor (inherits from MedicalStaff)

Additional Data Members:

- specialization
- consultationFee
- experience (in years)

Member Functions:

- acceptDoctorDetails()
- displayDoctorDetails()
- Calculate yearly income (consultationFee × estimated patients per month × 12)

Create Derived Class Nurse (inherits from MedicalStaff)

Additional Data Members:

- shift (Day/Night)
- wardAssigned
- salary

Member Functions:

- acceptNurseDetails()
- displayNurseDetails()

Create Derived Class LabTechnician (inherits from MedicalStaff)

Additional Data Members:

- labType
- numberOfTestsHandledPerDay
- salary

Member Functions:

- acceptLabTechnicianDetails()
- displayLabTechnicianDetails()

In main() Function

• Create one object each of:

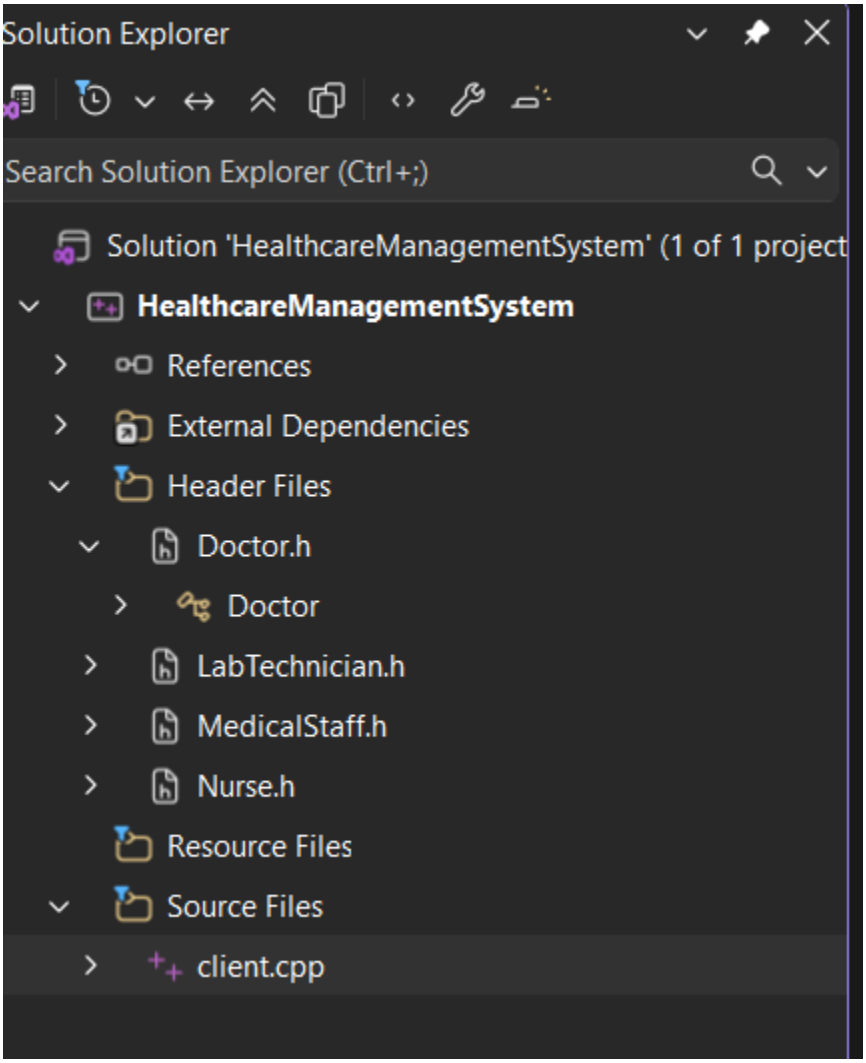
o Doctor

o Nurse

o LabTechnician

• Accept and display their complete details.

Structure Of Program



Solution: Assignment 2

Project Location:

D:\LearnC++\Day5_02032026\Ass2\ HealthcareManagementSystem

File Locations

Header File:

D:\LearnC++\Day5_02032026\Ass2\ HealthcareManagementSystem\MedicalStaff.h

```
#pragma once
#include<iostream>
using namespace std;
class MedicalStaff
{
private:
	int staffID;
	string name;
	int age;
	long long phoneNumber;
public:
	void acceptStaffDetails()
	{
		cout << "Enter staff ID: ";
		cin >> staffID;
		cout << "Enter name: ";
		cin >> name;
		cout << "Enter age: ";
```

```
		cin >> age;
		cout << "Enter phone number: ";
		cin >> phoneNumber;
	}
	void displayStaffDetails()
	{
		cout << "Staff ID: " << staffID << endl;
		cout << "Name: " << name << endl;
		cout << "Age: " << age << endl;
		cout << "Phone Number: " << phoneNumber << endl;
	}
};
```

D:\LearnC++\Day5_02032026\Ass2\ HealthcareManagementSystem\Doctor.h

```
#pragma once
#include "MedicalStaff.h"

class Doctor : public MedicalStaff
{
	private:
		string specialization;
		float consultationFee;
		float experienceYears;

	public:
		Doctor() {
```

```
        specialization = "";
        consultationFee = 0.0;
        experienceYears = 0.0;
}
void acceptDoctorDetails()
{
        acceptStaffDetails();
        cout << "Enter specialization: ";
        cin >> specialization;
        cout << "Enter consultation fee: ";
        cin >> consultationFee;
        cout << "Enter years of experience: ";
        cin >> experienceYears;
}
void displayDoctorDetails()
{
        displayStaffDetails();
        cout << "Specialization: " << specialization << endl;
        cout << "Consultation Fee: " << consultationFee << endl;
        cout << "Years of Experience: " << experienceYears << endl;
}
string getSpecialization()
{
        return specialization;
}
float getConsultationFee()
{
```

```
		return consultationFee;
	}
	float getExperienceYears()
	{
		return experienceYears;
	}

	void calculateYearlyIncome() {
		int estimatedPatientsPerMonth;
		cout << "Enter estimated number of patients per month: ";
		cin >> estimatedPatientsPerMonth;

		float income = consultationFee * estimatedPatientsPerMonth * 12; // 12 months
in a year
		cout << "Estimated Yearly Income: " << income << endl;
	}



};
```

D:\LearnC++\Day5_02032026\Ass2\ HealthcareManagementSystem\Nurse.h

```
#pragma once
#include<iostream>
#include"MedicalStaff.h"

class Nurse:public MedicalStaff{
	private:
```

```
        string shift;
        string wardAssigned;
        double salary;

public:
        Nurse() {
                shift = "";
                wardAssigned = " ";
                salary = 0.0;
        }
        Nurse(string shift, string wardAssigned, double salary) {
                this->shift = shift;
                this->wardAssigned = wardAssigned;
                this->salary = salary;
        }
        void acceptNurseDetails() {
                acceptStaffDetails();
                cout << "Enter shift (e.g., Day, Night): ";
                cin >> shift;
                cout << "Enter ward assigned: ";
                cin >> wardAssigned;
                cout << "Enter salary: ";
                cin >> salary;
        }
        void displayNurseDetails() {
                displayStaffDetails();
                cout << "Shift: " << shift << endl;
```

```
                cout << "Ward Assigned: " << wardAssigned << endl;
                cout << "Salary: " << salary << endl;
            }
};
```

D:\LearnC++\Day5_02032026\Ass2\ HealthcareManagementSystem\LabTechnician.h

```
#pragma once
#include<iostream>
#include"MedicalStaff.h"

class LabTechnician:public MedicalStaff
{
private:
    string labType;
    int numberofTestHandledPerDay;
    float salary;
public:
    LabTechnician()
    {
        labType = "";
        numberofTestHandledPerDay = 0;
        salary = 0.0;
    }
    LabTechnician(string labType, int numberofTestHandledPerDay, float salary)
    {
```

```
        this->labType = labType;
        this->numberofTestHandledPerDay = numberofTestHandledPerDay;
        this->salary = salary;
    }
    void acceptLabTechnicianDetails()
    {
        acceptStaffDetails();
        cout << "Enter lab type (e.g., Pathology, Radiology): ";
        cin >> labType;
        cout << "Enter number of tests handled per day: ";
        cin >> numberofTestHandledPerDay;
        cout << "Enter salary: ";
        cin >> salary;
    }
    void displayLabTechnicianDetails()
    {
        displayStaffDetails();
        cout << "Lab Type: " << labType << endl;
        cout << "Number of Tests Handled Per Day: " << numberofTestHandledPerDay << endl;
        cout << "Salary: " << salary << endl;
    }

};
```

Client/Main File

D:\LearnC++\Day5_02032026\Ass2\ HealthcareManagementSystem\Client.cpp

```
#include <iostream>
using namespace std;
#include "Doctor.h"
#include "Nurse.h"
#include "LabTechnician.h"

int main() {

    Doctor doctor;
    Nurse nurse;
    LabTechnician labTechnician;

    cout << "Enter details for Doctor:" << endl;
    doctor.acceptDoctorDetails();

    cout << "\nEnter details for Nurse:" << endl;
    nurse.acceptNurseDetails();

    cout << "\nEnter details for Lab Technician:" << endl;
    labTechnician.acceptLabTechnicianDetails();

    cout << "\nDetails of Doctor:" << endl;
    doctor.displayDoctorDetails();
```

```
    cout << "\nDetails of Nurse:" << endl;
    nurse.displayNurseDetails();

    cout << "\nDetails of Lab Technician:" << endl;
    labTechnician.displayLabTechnicianDetails();

    return 0;
}
```

OUTPUT

```
Microsoft Visual Studio Debug Console
Enter details for Doctor:
Enter staff ID: 1
Enter name: Vikas
Enter age: 35
Enter phone number: 8585747496
Enter specialization: ENT
Enter consultation fee: 500
Enter years of experience: 12

Enter details for Nurse:
Enter staff ID: 102
Enter name: Sanjana
Enter age: 26
Enter phone number: 7474747474
Enter shift (e.g., Day, Night): Day
Enter ward assigned: 204
Enter salary: 30000

Enter details for Lab Technician:
Enter staff ID: 103
Enter name: Sanjay
Enter age: 28
Enter phone number: 1234567890
Enter lab type (e.g., Pathology, Radiology): Pathology
Enter number of tests handled per day: 6
Enter salary: 31000

Details of Doctor:
Staff ID: 1
Name: Vikas
Age: 35
Phone Number: 8585747496
Specialization: ENT
Consultation Fee: 500
Years of Experience: 12

Details of Nurse:
Staff ID: 102
Name: Sanjana
Age: 26
Phone Number: 7474747474
Shift: Day
Ward Assigned: 204
Salary: 30000

Details of Lab Technician:
Staff ID: 103
Name: Sanjay
Age: 28
Phone Number: 1234567890
Lab Type: Pathology
Number of Tests Handled Per Day: 6
Salary: 31000

D:\LearnC++\Day5_02032026\Ass2\HealthcareManagementSystem\x64\Debug\HealthcareManagementSystem.exe (process 12156) exited with code 0 (0x0).
Press any key to close this window . . .
```